These principles focus on defining clear rules for using data (by providing licenses), provenance information, and recommending standards that apply to the domain. Subsections 1a and 1b are essential components of a well-structured data product. Subsection 1c can be seen as recommendations for creating a data product.

If you are interested in more details about FAIR implementation, see the initiative site at www.go-fair.org.

5.6.5 Immutable

FAIR can be an excellent addition to the essential criteria for creating a data product. Another critical aspect to consider is the mutability of the data provided by the data product. Mutability can be troublesome. Conclusions drawn from the source data may be hard to confirm later if the source data changes over time.

If, for example, decisions about ingredient proportions in a drug have been made based on the analysis of scientific studies, then getting to that unmodified source data may be necessary from a regulatory perspective. Therefore, data immutability should be a desirable characteristic of a data product.

DO DATA PRODUCTS ALWAYS HAVE TO BE IMMUTABLE?

One of the primary goals of introducing a data mesh is to democratize data by enabling access to the various data created within an organization. Not all data created in organizations is created with the same consumption patterns in mind (consumption of data from any point in the past). Yet, the value of making it available externally can be significant.

Despite the lack of immutability, data should be generally available as it is, especially in the context of data generated by transactional systems (e.g., read-only APIs based on command query responsibility segregation, or CQRS, architecture). Therefore, we treat immutability as a desirable feature of a data product. However, it is not a necessary condition that every data product must fulfill. A data product should explicitly declare whether its data is immutable and how past data can be accessed.